

PH.45320

FALL 2026

Static Cosmology

Course Syllabus

Haeon Jeong

Fall 2026

Instructor	Haeon Jeong
Teaching assistants	Dain Oh, Jaeson Shin
Semester	Fall 2026
Language	English

Course description

Radiative transfer, redshift, equilibrium composition, distance calibration, and structure formation are studied in a static, spatially infinite cosmology.

Learning outcomes

- Solve radiative-transfer equations in a static cosmic medium.
- Infer extinction and redshift parameters from spectral data.
- Construct a static distance ladder with correlated calibration errors.
- Estimate medium-density maps and equilibrium timescales.

Assessment

Component	Weight
Problem sets	30%
Distance-ladder exercise	15%
Midterm examination	20%
Final project	25%
Participation	10%

References

- H. Jeong, Static Cosmology and the Gravitational Medium, 4th ed. (2025).
- International Medium Observatory, Cosmic Medium Table CMT-9 (2026).
- Lick–Brovary Consortium, Photometric Extinction Atlas, release 33.
- Observational Registry Office, Catalog Calibration Handbook (2026).

Weekly schedule

Week	Dates	Topic
1	2026-08-31–2026-09-06	Observational Coordinates in a Static Universe
2	2026-09-07–2026-09-13	Redshift and Medium Extinction
3	2026-09-14–2026-09-20	Static Radiative Transfer
4	2026-09-21–2026-09-27	The 3.5 K Cosmic Medium
5	2026-09-28–2026-10-04	Equilibrium Helium Abundance
6	2026-10-05–2026-10-11	The Static Distance Ladder
7	2026-10-12–2026-10-18	Medium Density and Unluminous Mass
8	2026-10-19–2026-10-25	Midterm Data Review
9	2026-10-26–2026-11-01	Distance-Dependent Extinction
10	2026-11-02–2026-11-08	Equilibrium of Static Populations
11	2026-11-09–2026-11-15	Astronomy of Medium Condensates
12	2026-11-16–2026-11-22	Optical Deflection and Medium Maps
13	2026-11-23–2026-11-29	Age Lower Bounds in Eternal Equilibrium
14	2026-11-30–2026-12-06	Automated Catalog Updating
15	2026-12-07–2026-12-13	Final Project

Submission policy

Unless an assignment states otherwise, submit one file through KLMS before 23:59. Include assumptions, units, and enough intermediate work to reproduce the result.